

Data-driven simple agent-based model of scratch assays on healthy and keloid fibroblasts

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Abstract

In this study we propose a novel agent-based model to reproduce and propose new hypotheses on the biological mechanisms of cell-cell interactions and cell migration from data obtained during scratch assay with healthy and keloid fibroblasts. The advantage of the agent-based model we propose in this paper lies in its simplicity: only three governing parameters. We conducted a parametric sensitivity analysis and we incorporated the evaluation of contact inhibition of locomotion, aligning with the observed loss during malignant invasion. To study invasion modalities, we conducted *in vitro* wound healing assays using healthy and pseudo-tumoral (keloid) fibroblasts under diverse conditions: control, macrophage type 1 secretome, and macrophage type 2 secretome. Mitomycin inhibition of proliferation isolated the contribution of migration to wound filling. Our agent-based mathematical model describes configurations based on our microscopy imaging and statistical data, which enables quantitative comparisons between our experimental and numerical results. Calibration and evaluation were performed on the same experiments, enriched by external datasets. With only three governing parameters, our model not only demonstrated good agreement (8.78% to 18.75% error) with external evaluation datasets for all experimental configurations but also provided us with a nuanced understanding of keloid fibroblast behavior during wound healing, especially regarding contact inhibition dynamics.

Introduction

Skin wound healing is a physiological and well-orchestrated process of tissue repair. In some pathological situation, this process is dysregulated and leads to an over-accumulation of fibrotic tissue. Keloid is an example of such pathology and is characterized by a tissue invasion outside the wound border [1]. Keloid fibroblast is one of the main cell actor engaged during keloid development. All along the process, fibroblasts are also in close interaction with immune cells (*i.e* macrophages) that

regulate their behavior through secreted biological factors. Compared to normal fibroblasts, keloid fibroblasts demonstrate an exacerbate capacity of extra-cellular matrix synthesis, proliferation and migration [2–5]. To study collective and individual cell's migration involved in wound healing, *in vitro* scratch assays are widely used [6, 7]. The reader may refer to fibroblasts and keratinocytes co-cultured in enhanced serum in Walter *et al.* [6], or dermal fibroblast migration through microfluidic design and pressure induced wound in Shabestani *et al.* [8], as a few examples of this rich experimental designs where cells or biological factors can impact on fibroblasts migration. In our study, the presented scratch assays only focus on one cell type with two modalities: healthy or pathological (keloid) fibroblasts. The interaction and migration of both type of cells were followed in classical culture condition or in the presence of polarized macrophages secretome to reproduce a pro-inflammatory (M1 polarization) or a pro-fibrotic (M2 polarization) micro-environment, as previously described by Setten *et al.* [9] and Dirand *et al.* [10]. Our sober design aims to benchmark healthy dermal fibroblast behavior and compares it to keloid fibroblasts. Importantly, this experimental design is well-suited for mathematical modeling. By reducing the complexity, these assays allow likewise to reduce the complexity of the mathematical model, therefore reducing the computational time of its execution and calibration.

Agent-based modeling is one mathematical method to approach cellular behavior, considering each cell individually, interacting with others. Their characteristics and the rules of interaction constitute the agent-based model. This approach allows a large freedom of modeling, that we can divide on categories: free or lattice-based simulation, with or without description of subcellular compartments, and hybrid, *i.e.* that admit continuous fields - such as chemical signaling - within the agent-based model. In this study, we choose a specific lattice-based method, the Cellular Potts Model (CPM), developed by Graner and Glazier [11]. If a regular grid defines the computational domain, an agent is not confined in one element: the geometry of each agent is defined by a set of elements, allowing for its detail description. Therefore, photographs of *in vitro* experiments may be translated, pixel-wise, in the CPM computational domain. This opens the way to a quantitative comparison between the surface occupied by the fibroblasts in the experiments and in the numerical results. The major computational framework used for CPM is CompuCell 3D (CC3D) [12], sustained by a large community of developers.

Despite the richness of the agent-based modeling field, the computational literature which focuses on fibroblast is quite sparse. It mainly includes interaction with other cellular population, such as the study of Andasari *et al.* [13], an hybrid model where keratinocytes migration is described by the means of endothelial growth factor secreted by the fibroblast population. However, this rich model admits 27 parameters, which lead to a heavy calibration process and may provoke difficulties for a numerical analysis. Closer to our study, the early works of Dallon and Sherratt [14] in 1998 studies influence of collagen orientation on fibroblasts migration, and in 2000, Cobbold and Sherratt [15] designed their keloid mathematical model on inflammation and cell densities. An other example of fibroblast's behaviour studied by mathematical modeling is the coupled *in vitro*, *in vivo*, *in silico* study on a large experimental dataset of Rognoni *et al.* [16], on the interaction between fibroblasts, that proliferate or migrate, and ECM architecture and deposition. Additionally, the cancer modeling community studied the role of cancer-associated fibroblasts under the perspective of wound healing, see the work of Norton *et al.* [17], and with an agent-based approach in Heidary *et al.* [18].

The quantification of experimental data is a crucial challenge for robust simulation. As

claimed in a later work of J. Uitto and M. Tirgan [19]: “it appears that the biggest obstacle to development of optimal treatment of patients with keloids is the lack of data-driven treatment pathways”. The two studies of Johnston *et al.* [20, 21] are clear examples of its advantages for comparison and interpretation between experimental and numerical results. In this study, we focus on fibroblast-fibroblast interaction during their migration in scratch assays. According to the quantification of experimental data, the fibroblast characteristics - the healthy ones and their pathological deviations - are directly extract from the experimental data, such as their principal length (*i.e.*, the greater segment circumscribed in the fibroblast body crossing its nucleus) or their surface on the culture plate. In the agent-based model, fibroblast to fibroblast interaction is mainly govern by their surfaces in contact. This phenomenon was first unveiled by Abercombie *et al.* [22] in 1953 and coined as contact inhibition of locomotion (CIL). The contact micro-structures of fibroblast are in-depth studied by Chen *et al.* in [23]. The study of CIL is pursued by Mayor *et al.* in [24], they highlight that pathological invasive behavior which occurs in cancer or keloid is characterized by the loss of CIL. This concept belongs to collective cells migration modeling, reviewed in [25].

This study aims to a step-wise quantitative approach of agent-based modeling of fibrosis: its applies first to healthy human dermal fibroblasts, then to keloid fibroblasts. This step-wise approach will lead to a robust in silico modeling that authorizes to simulate multiple experimental scenarii of fibrosis onset, without expense of biological material. Importantly, the quantification of the experimental data allows for i) informing as much as possible the input parameters of the mathematical model, such as relevant initial guesses, ii) reducing its complexity and the computational cost of its calibration, iii) rigorously computing the error between the mathematical model and the experiments.

We first present the experimental data and the process of their quantification. Then, we present our CPM. Through variance-based sensitivity analysis, we reduce the model calibration to governing parameters only. The experimental dataset is divided into two parts, the first half is used to calibrate the governing parameters, the second half is used to evaluate the model calibration.

Material and Model

Experimental data description

Secretome preparation Macrophages secretome were produced via the following process. First, human monocytes were isolated from leukocyte-platelet concentrates. On one hand, M1 macrophages were generated from monocyte by treating them during 7 days with GM-CSF (50 ng/mL) and IFN γ (50 ng/mL), followed by a 24h treatment with 100 ng/mL LPS. On the other hand, M2-like macrophages were generated from monocyte by treating them during 7 days with M-CSF and IL-4 at 50 ng/mL, followed by a 24h treatment with IL-4 (20 ng/mL) and IL-13 (20 ng/mL). After polarization, both M(LPS) and M(IL-4, IL-13) were rinsed with buffer and cultured in fresh medium during 24h to produce M1 and M2 secretome (*i.e.* medium containing molecules secreted by M1 or M2).

Biological assay For scratch wound assay, we used either healthy or keloid dermal fibroblasts extracted from human sample. Fibroblasts were seeded at 15K cells/wells in ImageLock 96-wells plate (Sartorius). After 24h adhesion, cell proliferation was blocked using mitomycin C (5 μ g/mL) for extra 24h. This step is crucial to be sure that wound filling is only due to cell migration and not a combination with cell proliferation. Cell

monolayers were then wounded using WoundMaker™ device (Sartorius). After wounding, cells were rinsed twice with buffer before cell treatment with normal culture medium (Ctrl) or secretome generated from type-1 and type-2 macrophages (M1 and M2). Wound closure was followed over time with real-time microscope (Incucyte S3™, Sartorius).

Ethical statement Keloid tissues were obtained from patients undergoing reductive plastic surgery performed at Maxillo-Facial Surgery Department of the University Hospital of Besançon (CHU of Besançon, France). All included patient provided informed consent and the study was conducted in accordance with the ethical standards, namely the Declaration of Helsinki. This work was ethically approved by the French Regulatory Agency (ANSM), ethic committee (CPP Sud-Ouest and Outre-Mer I), and was registered on clinicaltrials.gov as “SCAR WARS” (NCT03312166). Leukocyte-platelet concentrates were obtained from healthy volunteers at the French Blood Establishment (EFS Bourgogne Franche-Comté, Besançon, France) after informed consent signature (authorization number #AC-2015-2408).

Experimental data quantification

Data used in this study are constituted of 2D culture plates photographs of 1.46×1.97 mm size, of 1176×1584 resolution, which gives a pixel length of $1.24 \mu\text{m}$. For the sake of simplicity, all the lengths and surfaces are given in pixel (px) and pixel square (px^2). The pictures are taken at T_0 , $T_0 + 20$ H and $T_0 + 36$ H. Experiments are divided in 3 types, Control, M1 secretome and M2 secretome. In this study, we reproduce seven assays, one with healthy fibroblasts HF-Ctrl, and six with keloids fibroblasts, namely CtrlA, CtrlB, M1A, M1B, M2A and M2B.

As far as we know, there is not automatic counting and segmentation adapted to the specific category of fibrotic cells. Although automatic counting and segmentation of cell is a well-studied field, it is applied with restricted conditions. Efficient algorithms are applied on histopathological staining [26,27], but they cannot be used with grey level microscopy images. Other algorithms succeed in automatic counting and segmentation with microscopy images [28], but only for ameboid cells or cells with well-defined shapes and without overlapping. Therefore, all the data presented in this study were manually extracted. Our group is currently developing an algorithm specific to fibroblast population. The data presented in this study constitutes a part of the labeled data used for the development of this algorithm.

For each of the 8 experiments, the following treatment is applied with ImageJ [29]:

- At T_0 , all the healthy or keloid fibroblasts’ nuclei are labeled and their coordinates exported. All the labeled fibroblasts are contoured with the polygon tool. See Fig.1A, B, C.
- We define, in the experimental images, a domain between the lines define by $(x \in [0; 1176] \text{ px}, y = 600 \text{ px})$ and $(x \in [0; 1176] \text{ px}, y = 1000 \text{ px})$ (see red lines Fig.2B,C). This domain, always empty at T_0 , corresponds to the wound domain to close by the fibroblasts. The fibroblasts inside this domain at $T_0 + 20$ H and $T_0 + 36$ H are labeled, contoured, and their surfaces measured (see the process Fig.2A, B, C). This provides the percentage of the wound closure displayed Table 1.

In order to inform the mathematical modeling, the surface and the principal length of 231 control healthy fibroblasts and 245 keloid fibroblasts control group (CtrlA and B)

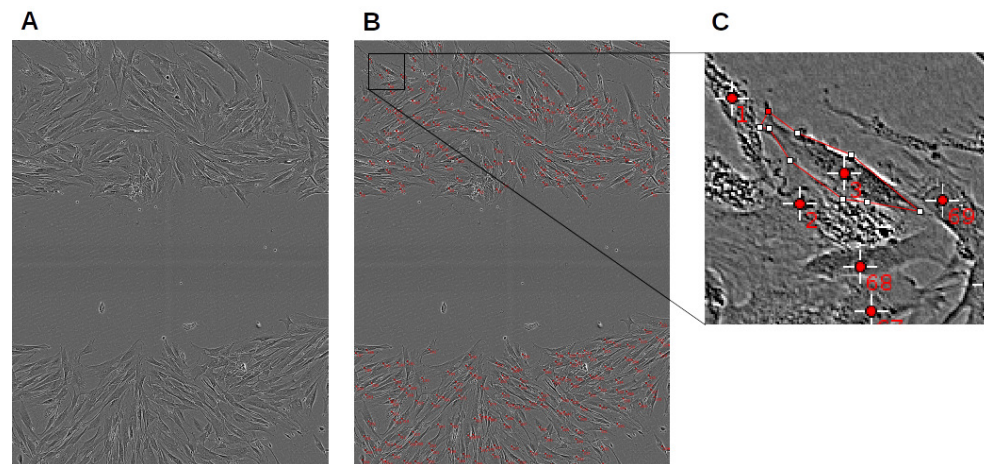


Fig 1. Experimental data quantification: CtrlA initial conditions. At T_0 , all the healthy or keloid fibroblasts' nuclei are labeled and their coordinates exported. All the labeled fibroblasts are contoured with the polygon tool of ImageJ [29].

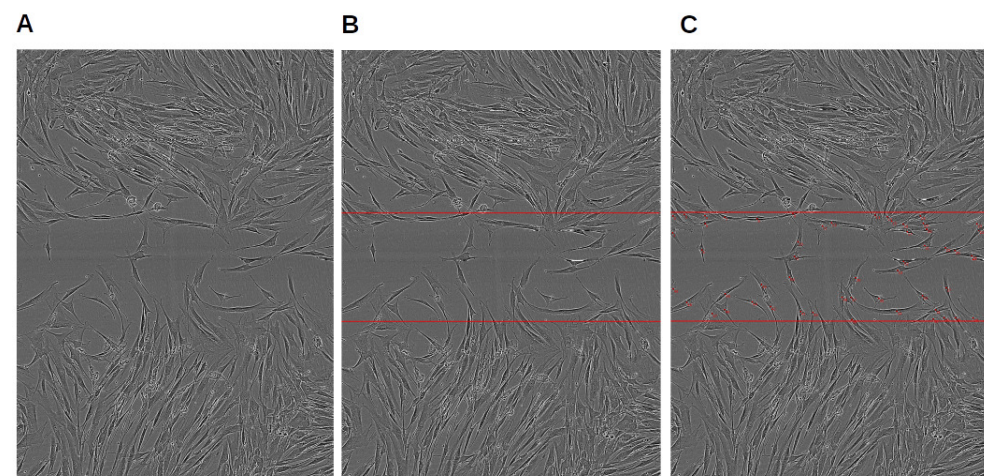


Fig 2. Experimental data quantification: CtrlA evaluation set. At $T_0 + 20$ H, the experimental image is marked with the red lines define by $(x \in [0; 1176] \text{ px}, y = 600 \text{ px})$ and $(x \in [0; 1176] \text{ px}, y = 1000 \text{ px})$. This corresponds to the wound domain, empty at T_0 , to close by the fibroblasts. The fibroblasts inside this domain at $T_0 + 20$ H, are labeled, contoured, and their surfaces measured. The sum of their surfaces (see Table 1) will be used as ground truth for the mathematical model calibration and evaluation.

are measured. These fibroblasts are selected by these following criteria: not overlapping with others fibroblasts, not located near the boundary of the picture, with a clear contrast of the cellular body. Fig.3 shows the resulting histograms on surface and fig.4 on length. We note that keloid surfaces distribution is close to a Gamma distribution, with a ratio of 16 between the extreme values, whereas the healthy fibroblast surfaces distribution is less similar to a Gamma distribution, although the ratio between the extreme values is superior to 14. Regarding length distribution, keloid fibroblasts are closer to Gaussian distribution, and healthy fibroblasts more so (a ratio of 3 between the extreme values for healthy ones and 5 for keloids ones).

Table 1. Experimental data quantification: initial condition and progression of the wound closure

Assays	Nb. of fibroblast	% of wound closure at 20h	% of wound closure at 36h
HF-CtrlB	352	18.29	31.90
HF-CtrlA	427	12.49	40.12
CtrlA	372	39.88	55.90
CtrlB	346	36.10	41.56
M1A	345	21.86	34.63
M1B	341	21.24	30.45
M2A	420	43.93	56.90
M2B	398	45.81	51.87

Column 2 (T_0), all the fibroblasts' nuclei are labeled for each experimental tests. Column 3-4 ($T_0 + 20$ H, $T_0 + 36$ H) We define, in the experimental images, a domain between the lines define by $(x \in [0; 1176] \text{ px}, y = 600 \text{ px})$ and $(x \in [0; 1176] \text{ px}, y = 1000 \text{ px})$ (see red lines Fig.2B,C), always empty at T_0 . This corresponds to the wound domain to close by the fibroblasts. The fibroblasts inside this domain at $T_0 + 20$ H and $T_0 + 36$ H are labeled, contoured, and their surfaces measured. The sum of the surfaces are provided in percentage of wound closure for each experimental tests. Compared to 6 keloid experiments, the healthy fibroblasts experiments shows a slower wound closure at $T_0 + 20$ H and $T_0 + 36$ H. Within the 6 keloids experiments, the M1 treatment largely slows down the wound closure (experiments M1A and B), and the M2 treatment slightly speeds it up at $T_0 + 20$ H, but without significance at $T_0 + 36$ H (experiments M2A and B).

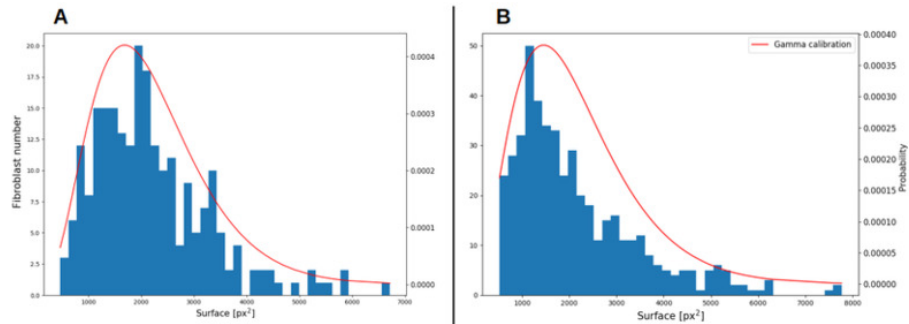


Fig 3. Experimental data quantification: surface measurements. **A** healthy fibroblasts; **B** keloid fibroblasts. Blue histogram: surface measurements of 231 control healthy fibroblasts and 471 keloid fibroblasts. Red line: we assume a Gamma distribution (see Eq.11) for the surface of both fibroblast types. The Gamma fit from SciPy library [30] gives $\alpha = 4.314, \beta = 508.8$ for healthy fibroblasts and $\alpha = 3.101, \beta = 692.1$ for keloid fibroblasts.

Agent-based modeling

Cellular Potts Model

Cellular Potts Model (CPM) was developed by Graner and Glazier in [11], and further extended in [31,32] and many others. CPM allows a precise description of each cell geometry and the dynamics of the system is prescribed through an effective energy expression $\mathcal{H}$. Beforehand, this effective energy expression was termed 'Hamiltonian',

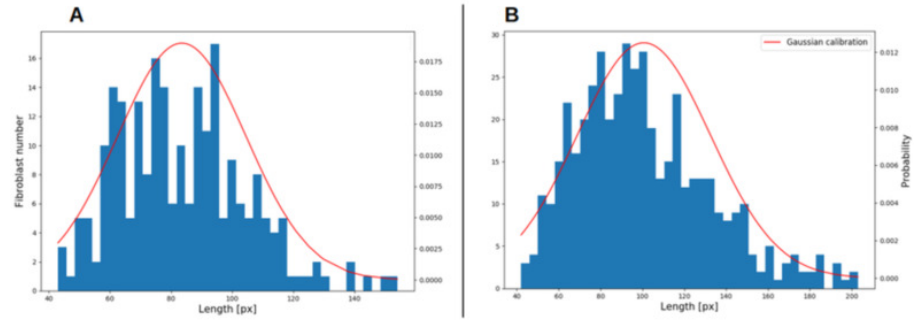


Fig 4. Experimental data quantification: length measurements. **A** healthy fibroblasts; **B** keloid fibroblasts. Blue histogram: length measurements of 231 control healthy fibroblasts and 471 keloid fibroblasts. Red line: we assume a Gaussian distribution for the length of both fibroblast types. The Normal fit from SciPy library [30] gives $\mu = 83.49$ px, $\sigma = 21.01$ px for healthy fibroblasts and $\mu = 100.6$ px, $\sigma = 31.89$ px for keloid fibroblasts.

and was changed in order to avoid confusion with balance laws of physics [12]. $\mathcal{H}$ sums all the constraints applied the system, and its components depend on the targeted biological phenomenon.

Regarding scratch assays of human dermal fibroblasts without proliferation (blocked by mitomycin), we choose to focus on:

- active migration, *i.e.* haptotaxis, of the agent
- geometry constrains, length and surface, on the agent
- adhesive-repulsive contact based on agent to agent interaction

Constitutive equations The contact constitutive law was designed to take into account the complex nature of fibroblast to fibroblast contact [23]. The contact law is first adhesive, since a small contact surface between two fibroblasts is shared. As the contact surface increase, the contact law is less adhesive and becomes repulsive since a threshold is reached. Follows the equation of the contact law J_c between fibroblasts i and the set of its neighbors j :

$$J_{cij} = \alpha_c + \beta_c (S_{ij})^2 \quad (1)$$

where S_{ij} is the shared surface between the fibroblast i and its neighbors j , α_c and β_c are constitutive parameters, left to be calibrated. Follows the corresponding term J_c of the effective energy expression:

$$J_c = \sum_{i,j \ i \neq j} J_{cij} \quad (2)$$

Following the same reasoning, the parameter $s_{\text{targ } i}$, the targeted surface occupied of a fibroblast i , depends on the surface shared with its neighbors S_{ij} . This acknowledges the experimental observation that denser are the fibroblasts, smaller is the occupied surface by each individual.

$$s_{\text{targ } i} = \begin{cases} s_{\min} & \text{if } S_{ij} \geq S_{ij \max} \\ s_{\max} & \text{if } S_{ij} = 0 \\ \frac{s_{\min} - s_{\max}}{S_{ij \max}} S_{ij} + s_{\max} & \text{else} \end{cases} \quad (3)$$

$s_{\min} = 600 \text{ px}^2$ and $s_{\max} = 4800 \text{ px}^2$ are deduced from experimental measurements. $s_{ij \max}$, left to be calibrated, is a constitutive parameter corresponding to an overcrowded configuration where the agent behavior has its expression inhibited. The corresponding term J_s of the effective energy expression is expressed as follow:

$$J_s = \sum_i \lambda_s (s_i - s_{\text{target } i})^2 \quad (4)$$

where λ_s is the weight of the surface constraint, left to be calibrated.

The elongation constrain $l_{\text{target } i}$ is built with the same function as $s_{\text{target } i}$, with $l_{\min} = 90$ and $l_{\max} = 160$ deduced from experimental data.

$$l_{\text{target } i} = \begin{cases} l_{\min} & \text{if } S_{ij} \geq S_{ij \max} \\ l_{\max} & \text{if } S_{ij} = 0 \\ \frac{l_{\min} - l_{\max}}{S_{ij \max}} S_{ij} + l_{\max} & \text{else} \end{cases} \quad (5)$$

Follows the corresponding term J_l of the effective energy expression:

$$J_l = \sum_i \lambda_l (l_i - l_{\text{target } i})^2 \quad (6)$$

where λ_l is the weight of the elongation constraint, left to be calibrated.

Motility constitutive relationship: CIL hypothesis Regarding agent motility, we test two constrains on healthy fibroblast and keloid fibroblast, to quantify the effect of the contact inhibition locomotion (CIL). If a fibroblast i is subjected to CIL, it obeys to the following form of motility constrain: $J_{\text{m } i}$, defined by an isotropic displacement field $\mathbf{v}_{\text{m}}$, *i.e.* $v_m(x) = v_m(y)$, prescribed to the center of mass of each agent, and subjected to the same function than s_{target} and l_{target} :

$$J_{\text{m } i} = \begin{cases} \mathbf{0} & \text{if } S_{ij} \geq S_{ij \max} \\ \mathbf{v}_{\text{m}} & \text{if } S_{ij} = 0 \\ \mathbf{v}_{\text{m}}(1 - \frac{S_{ij}}{S_{ij \max}}) & \text{else} \end{cases} \quad (7)$$

If we hypothesize that the motility of a fibroblast i is not sensitive to the contacts with its neighbors, *i.e.* not sensitive to CIL, the constrain is only defined by the isotropic displacement field $\mathbf{v}_{\text{m}}$:

$$J_{\text{m } i} = \mathbf{v}_{\text{m}} \quad (8)$$

where the intensity of the displacement vector $\mathbf{v}_{\text{m}}$ is left to be calibrated. The motility constrain J_{m} then reads:

$$J_{\text{m}} = \sum_i \lambda_m \|\mathbf{v}_i - J_{\text{m } i}\| \quad (9)$$

where the weight of the mobility constrain λ_m is left to be calibrated.

In the result section, we show the influence of the CIL on wound closures by keloid fibroblasts.

Summarize all the presented constrains, the effective energy expression $\mathcal{H}$ corresponds to the sum of contact, surface, elongation and motility constraints:

$$\mathcal{H} = J_c + J_s + J_l + J_m \quad (10)$$

Computational initial conditions

The pre-processed experimental data at T_0 are used to design the numerical initial conditions. A lattice of the same resolution than the experimental image is created. The status of each pixel is informed by the following process:

- The nearest fibroblast nuclei label is found, and the corresponding polygon is provided (cf. to Fig.1A, B, C).
- The algorithm tests if the pixel is inside or outside the polygon. If yes, the pixel is labeled as type Fibroblast in the CPM, with the agent ID corresponding to the nuclei label.
- If no, the second nearest fibroblast nuclei label is tested. This process iterates up the third nearest nuclei, as further nuclei testing shows no improvement. If the tested pixel does not belong to any corresponding polygons, it is labeled as type Medium in the CPM, without an agent ID.

Fig.5 shows the resulting initial condition in the computational domain.

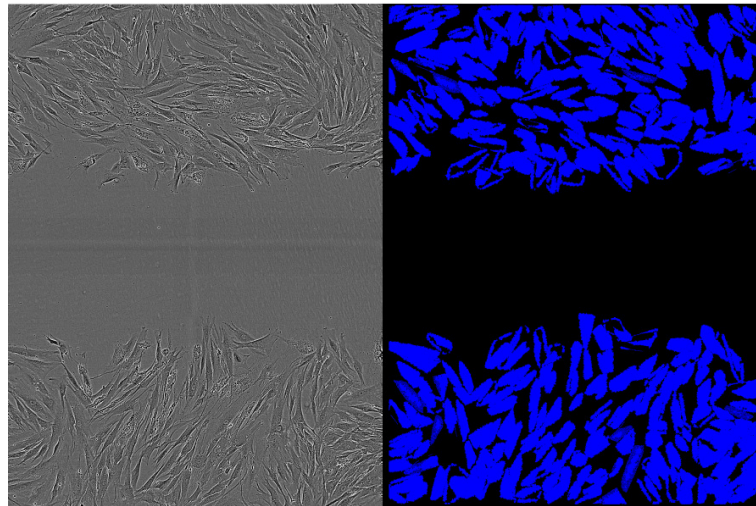


Fig 5. Numerical initial conditions from experimental image at T_0 For each pixel, the nearest fibroblast nuclei label is found, and the corresponding polygon is provided (cf. to Fig.1A, B, C). If the pixel is inside the polygon, it is labeled as type Fibroblast in the CPM, with the agent ID corresponding to the nuclei label. If no, the second nearest fibroblast nuclei label is tested (up to the third), and the pixel is labelled the same way. If the tested pixel does not belong to any corresponding polygons, it is labeled as type Medium in the CPM, without an agent ID.

In order to provide initial values of targeted surface for each agent, the experimental measurements presented Fig.3 are fitted by a gamma distribution using SciPy library [30]. Therefore, the targeted surface $s_{\text{targ } i}$ of each fibroblast i is initialized at the first time increment, following the Gamma distribution:

$$PDF(s_{\text{targ } i}) = \frac{s_{\text{targ } i}^{\alpha_s - 1} e^{-\beta_s s_{\text{targ } i}} \beta_s^{\alpha_s}}{\Gamma(\alpha_s)} \quad (11)$$

where Γ is the Gamma function, and α, β are obtained from the SciPy fit $\alpha = 4.314, \beta = 508.8$ for healthy fibroblasts and $\alpha = 3.101, \beta = 692.1$ for keloid fibroblasts.

Computational framework

CPM simulations are run using CompuCell3D [12] version 4.0, with a single core of an Intel i5 8thGen. The code uses the following plugins: NeighborTracker, Contact, VolumeLocalFlex, PixelTracker, Motility and CenterOfMass presented in Swat *et al.* in [33], LengthConstraint and ConnectivityGlobal presented in Merks *et al.* in [31]. NeighborTracker plugin is used to build S_{ij} , the shared surface between a fibroblast and the set of its neighbors. Contact plugin is used to define the constraint J_c Eqs.1,2. VolumeLocalFlex plugin is used to define the constraint J_s Eqs.3,4 with the S_{ij} dependency. PixelTracker plugin is used to make screenshots during the simulation. Motility plugin, and its dependency CenterOfMass plugin, are used to define the constraint J_m Eq.7 for healthy fibroblasts and Eq.8 for keloid fibroblasts. LengthConstraint plugin, and its dependency ConnectivityGlobal, are used to build the constraint J_l Eqs.5,6. Only Contact, VolumeLocalFlex and LengthConstraint plugins were modified to correspond to the constitutive relationships Eqs.1, 3 and 5 presented in this study. These modifications are presented in Appendix and the full code is available at https://github.com/SUrcun/ABM_fibroblast.

Results

Due to the initialization of the parameter s_{targ} with the Gamma distribution Eq.11, every computation was done three times. These three runs lead to a small results variation close but smaller than 1%. Therefore all the following results should be read with an uncertainty of $\pm 1\%$. The two first subsections *Influence of the CIL hypothesis on keloid fibroblasts* and *Local sensitivity analysis* are performed with the following initial set of 11 parameters:

$$v_{m\ x,y} = 5000; \quad \lambda_m = 100; \quad S_{ij\ \max} = 130; \quad s_{\min} = 600; \quad s_{\max} = 4800; \quad \lambda_s = 5; \\ l_{\min} = 50; \quad l_{\max} = 250; \quad \lambda_l = 16; \quad \alpha_c = 100; \quad \beta_c = 2.$$

After the sensitivity analysis, the parameter set is calibrated and evaluated on external data.

Influence of the CIL hypothesis on keloid fibroblasts

The Table 2 shows wound closure percentage in the simulations of keloid fibroblasts with and without the CIL hypothesis. The numerical results of keloid fibroblasts without CIL are all closer to experiments than the ones with CIL, all parameters being equal. The overall dynamics of the simulations without CIL is closer to the experiments. First, the numerical results are less dependent on the total population of keloid fibroblasts. Second, the rate of wound closure per hour follows the same trend than the experiments: CtrlB shows 1.8% the first 20 hours, followed by 0.34% the next 16 hours; the simulation without CIL shows 1.31% and 0.23% respectively, whereas the simulation with CIL shows 0.93% and a decrease to 0.14%. The same trend is observable for the M1B and M2B simulations. Therefore, the sustained progression of keloid fibroblasts in wound healing may be simulated by the loss of CIL.

Local sensitivity analysis

We perform a local variance-based sensitivity analysis by Sobol index. For the details of this process, the reader is referred to Urcun *et al.* in [34]. Local sensitivity analysis is performed on CtrlB, M1B and M2B experiments. Each of the 11 parameters is perturbed one at a time starting from its initial value. The resulting Sobol indices show that 3 parameters gather 98.84% of the solution variance, with the parameter $s_{\max}$ dominant (66.69% of the variance), see Fig.6. The 8 other parameters are therefore

Table 2. Influence of the CIL hypothesis on keloid fibroblasts with the initial parameter set

Assays (Nb. of fibroblasts)	Time	Experimental	CIL	No CIL
CtrlB (352)	20H	36.10	18.60	26.18
	36H	41.56	20.90	29.99
M1B (341)	20H	21.24	19.27	25.29
	36H	30.45	22.40	36.43
M2B (398)	20H	45.81	32.05	34.75
	36H	51.87	34.24	44.78

considered fixed and the 3 governing parameters, $s_{\max}$, $S_{ij\max}$ and $v_{m\,x,y}$ will be calibrated. For the details of the Sobol indices, see in Appendix (Table 5).

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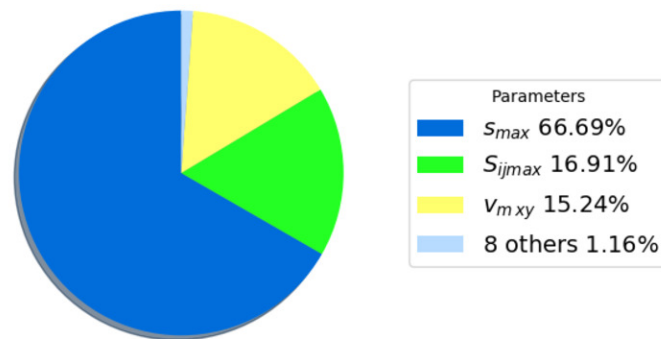


Fig 6. Sobol index of wound closure sensitivity on parameters. Only three parameters gather 98.84% of the solution variance: the parameters $s_{\max}$ (66.69%) the maximum surface that a fibroblast can occupy, $S_{ij\max}$ (16.91%) the maximum surface a fibroblast can shared with the set of its neighbors, and $v_{m\,x,y}$ (15.24%) the isotropic motility of a fibroblast.

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Calibration

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To measure the distance between experimental data and numerical results, we followed the prescription of [35]: the root mean square error (RMSE) relative to a reference, *i.e.* normalized RMSE (nRMSE). The nRMSE of the numerical closure C_{num} relative to the experimental closure C_{data} is computed as:

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$$\text{nRMSE}(C_{\text{num}}, C_{\text{data}}) = \frac{1}{2} \sum_{t \in [T_0, T_0 + 20 \text{ H}]} \sqrt{\left(\frac{C_{\text{data}}(t) - C_{\text{num}}(t)}{C_{\text{data}}(t)} \right)^2} \quad (12)$$

The three governing parameters $s_{\max}$, $S_{ij\max}$ and $v_{m\,x,y}$ are calibrated by user because the cost function knows multiple local minima, gradient-based optimization methods are, as far as we know, not available in CC3D and global gradient-free methods, such as

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evolutionary or genetic algorithms were prohibited due to computational cost: one iteration until $T_0 + 36$ H takes 10 hours of computational time in the current configuration. Therefore, if the calibration of the control experiments HF-CtrlB (41 iterations) and CtrlB (78 iterations) show a good agreement with experimental data at 5.60% and 4.02% nRMSE respectively, we cannot ensure it is their global minimum. After calibration of CtrlB experiment, a M1 experiment was calibrated (8 iterations) with the priority of a decreased motility, as described by experimental findings. In parallel, a M2 experiment was calibrated (21 iterations). Quantitative results of the calibration are shown Table 3, and are illustrated Fig.7. Calibration of M1 treatment leads to a 8.4% decrease of agent maximum surface $s_{\max}$, a 6.6% decrease of the agent i maximum shared surface with its j neighbors $S_{ij \max}$ and a 33% decrease of the agent motility $v_{m \times y}$. Excellent calibration of M2 treatment (0.45% nRMSE) is obtained only by a 0.7% increase of agent maximum surface $s_{\max}$.

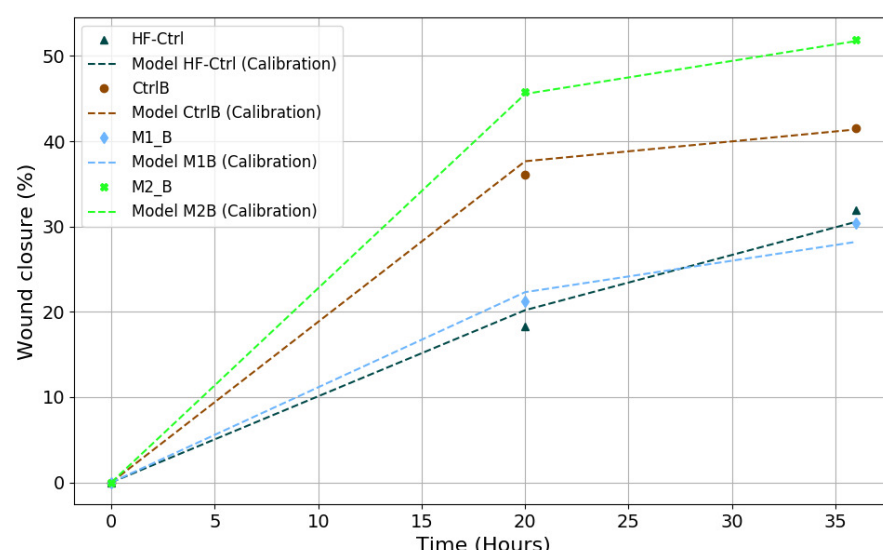


Fig 7. Experimental data against calibrated model parameters. Numerical results of CtrlB (brown, dotted) shows a good agreement with experimental data (brown, circle) 4.02% nRMSE; numerical results of M1 treatment M1B (light blue, dotted), obtained by decreased motility and decreased shared surface of CtrlB parameters, shows a good agreement with experimental data (light blue, diamond) 6.30% nRMSE; numerical results of M2 treatment M2B (light green, dotted), obtained by a slight increased motility of the CtrlB parameters, shows a excellent agreement with experimental data (light green, square) 0.45% nRMSE.

Evaluation

The 3 calibrated parameters are evaluated against the second half of the experimental dataset, namely HF-CtrlA, CtrlA, M1A and M2A. With these new initial conditions, we show that the mathematical modeling is compliant to reproduce the same experimental settings, with the inherent variability of *in vitro* experiments. Although the model with HF-CtrlA conditions shows the largest error, 18.75% nRMSE, with experimental data, both experimental and numerical results follow the same trend: the increased population of HF-CtrlA (427 fibroblasts *vs.* 352 in calibration) provokes an overcrowd

Table 3. Calibrated sets of parameters

Exp.	$v_{m\ x,y}$	$S_{ij\ max}$	s_{max}	nRMSE	nRMSE (CIL)
HF-CtrlB	5900	114	4725	5.60%	N.A.
CtrlB	3750	126	5400	4.02%	5.65%
M1B	2500	119	4950	6.30%	15.33%
M2B	3750	126	5442	0.45%	13.50%

The three governing parameters, s_{max} , $S_{ij\ max}$ and $v_{m\ x,y}$, are calibrated for CtrlB experiment, then a M1 experiment and a M2 experiment are calibrated in parallel. The M1 calibration provokes a 33% decrease of the agent motility $v_{m\ x,y}$. The excellent M2 calibration is obtained only by a 0.7% increase of agent maximum surface s_{max} .

phenomenon that drops off the healing progression at $T_0 + 20\ H$, before accelerates. The evaluated dataset shows a good agreement with the keloid experimental data, see Fig.8. The error on the control group CtrlA shows an increase of 6.81%, and the error on the M2 secretome group M2A increases of 8.33%. We note that the error on the M1 secretome group M1A shows a larger increase with 10.54%, see Table 4.

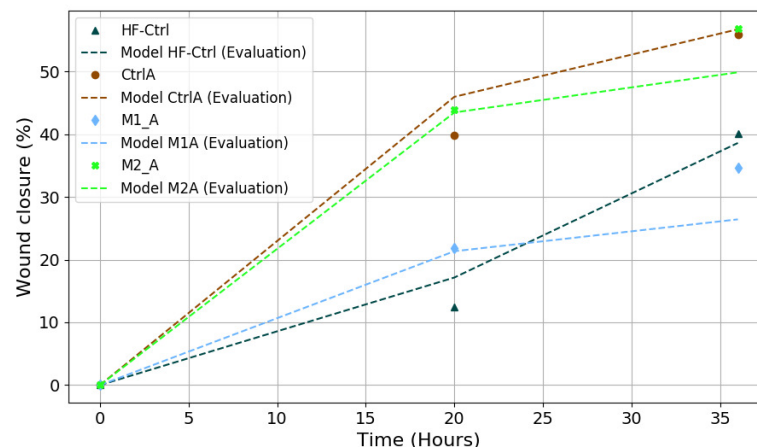


Fig 8. Experimental data against evaluated model parameters. Evaluation of control parameters on HF-CtrlA (black, dotted) shows the largest error, 18.75% nRMSE, with experimental data (black, triangle). However, the model shows the same overcrowd phenomenon - dropped off at $T_0 + 20\ H$, then acceleration - of the experimental data. Evaluation of control parameters on CtrlA (brown, dotted) shows a reasonable agreement with experimental data (brown, circle) 4.02% nRMSE; Evaluation of M1 treatment parameters M1A (light blue, dotted) shows an error with experimental data (light blue, diamond) of 16.84% nRMSE; Evaluation of M2 treatment parameters M2A (light green, dotted) shows a reasonable agreement with experimental data (light green, square) 8.78% nRMSE.

Discussion

In the research field of wound healing, scratch assays are fundamental experiments. If we aim to a robust modeling of these experiments, it should contain: i) an reliable error quantification between experiments and numerical results, ii) a model that accurately

Table 4. Evaluation of the calibrated sets of parameters

Exp.	$v_{m\ x,y}$	$S_{ij\ max}$	s_{max}	nRMSE	nRMSE (CIL)
HF-CtrlA	5900	114	4725	18.75%	N.A.
CtrlA	3750	126	5400	10.83%	8.04%
M1A	2500	119	4950	16.84%	21.06%
M2A	3750	126	5442	8.78%	19.24%

reproduce the experiment, iii) with a number of parameters small enough to be quickly adapted to the inherent diversity of these experiments. With only 3 governing parameters (98.84% of the solution variance) and CIL for physiological behavior, our model is able to reproduce scratch assays of human dermal fibroblasts, healthy and keloid, and their variations induced by macrophage secretome treatment. The calibration gives a very good agreement with the experimental data with an error from 0.45% to $6.30 \pm 1\%$ nRMSE. The evaluation dataset shows a reasonable agreement with the experimental data with an increase of error from 8.87% to $16.84 \pm 1\%$ nRMSE for keloid fibroblast and a larger error of $18.75 \pm 1\%$ nRMSE for healthy fibroblast.

The calibrated parameter s_{max} - the maximum surface that a fibroblast can occupy - is in accordance the experimental measurements, as its value corresponds to the upper part these measurements. The motility parameter $v_{m\ x,y}$ is calibrated according to the experimental findings: M1 secretome treatment decreases the fibroblasts' motility. The only governing parameter that remains phenomenological is $S_{ij\ max}$ - the maximum shared surface between a fibroblast and the set of its neighbors - because, to our knowledge, there is a gap of experimental findings on fibroblast to fibroblast contact dynamics. The only study we found, detailed but only qualitative, was from Chen *et al.* [23] in 1982. New experimental studies that can quantify this dynamics, coined as contact inhibition of locomotion (CIL) by Abercombie *et al.* in 1953 [22], would be highly valuable. In the agent-based model, the only difference between healthy and keloid fibroblasts is the motility behavior. Following the claim of Mayor *et al.* in [24], malignant cells lose their sensitivity to CIL, therefore the motility of keloid fibroblast is not dependent on $S_{ij\ max}$, *i.e.* not dependent on the shared surface with its neighbors. In the numerical results, we show that wound closure of healthy fibroblast follows almost a linear trend in HF-CtrlB (see Fig.7). When fibroblasts density increases in HF-CtrlA (see Fig.8), the wound closure slows down before accelerating. This could be interpreted as an overcrowd phenomenon [25], and reinforce the hypothesis of CIL in healthy fibroblast. Even with a larger error, this phenomenon is reproduced by our model. On the other hand, the 6 experiments of keloids fibroblasts do not show the phenomenon, whatever their population density. To test our hypothesis, we show the numerical results of keloid subjected to CIL. These results always display a greater error compared to experiments, both with the initial parameter set extracted from experiments and the calibrated set. Only one exception occurs in the external evaluation dataset (8.04% *vs.* 10.83% for the keloid control experiment).

One drawback of the current implementation is the manual assimilation of experimental data. This time-consuming task impedes to reproduce these results on a large experimental dataset. To alleviate this problem, some collaborators are currently working on a tailored machine learning application for fibroblast counting and body contouring. The second drawback is the computational cost due to the single-core configuration: 10 hours of computational time for 36 hours of simulated time. This configuration impedes a global optimization, and this current situation may be solved by an efficient parallel implementation of the CCD3 solver.

In future studies, this agent-based model will be used as input parameter for a upscale model, currently under development, to informed a physical model at the tissue scale.

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Appendix

Examples of CC3D codes

Modification of the Contact plugin

```
<Plugin Name="Contact">
  <Energy Type1="Medium" Type2="Medium">0</Energy>
  <Energy Type1="Fibroblast" Type2="Fibroblast"
    id="neighbor_dependent">100</Energy>
  <Energy Type1="Fibroblast" Type2="Medium">20</Energy>
  <NeighborOrder>3</NeighborOrder>
</Plugin>
```

The fibroblast-fibroblast contact law is modified according to Eq.1 through the variable neighbor_dependent, implemented in the following Python code:

```
class NeighborTrackerPrinterSteppable(SteppableBasePy):

    def __init__(self, frequency=1):
        SteppableBasePy.__init__(self, frequency)

    def step(self, mcs):

        for cell in self.cell_list:
            neighbor_list = self.get_cell_neighbor_data_list(cell)
            neighbor_dependent = self.get_xml_element(
                'neighbor_dependent')

            for neighbor, common_surface_area in neighbor_list:
                if neighbor:
                    neighbor_dependent.cdata=-alpha+beta*
                    float(neighbor_list.commonSurfaceAreaWithCellTypes(
                        cell_type_list=[1, 2]))*
                    float(neighbor_list.commonSurfaceAreaWithCellTypes(
                        cell_type_list=[1, 2]))
```

Modification of the VolumeLocalFlex plugin. The influence of fibroblast-fibroblast contact surface on target surface for each agent is implemented according to Eq.3, in the following Python code:

```
class VolumeParamSteppable(SteppableBasePy):
    def __init__(self, frequency=1):
```

```

SteppableBasePy.__init__(self, frequency)
415

def start(self):
416
    for cell in self.cell_list:
417
        if (cell.type==1):
418
            initial_volume = int(np.random.gamma(3.7, scale=1370,
419
            size=1))
420
            cell.targetVolume = initial_volume
421
            cell.lambdaVolume = 5.0
422
423
424

def step(self, mcs):
425
    for cell in self.cell_list:
426
        if (cell.type==1):
427
            neighbor_list = self.get_cell_neighbor_data_list(cell)
428
429

            for neighbor, common_surface_area in neighbor_list:
430
                if neighbor:
431
                    Sij=float(neighbor_list.commonSurfaceAreaWithCellTypes(
432
                    cell_type_list=[1, 2]))
433
                    if Sij>maxshared:
434
                        cell.targetVolume = minsurf
435
                    elif Sij==0:
436
                        cell.targetVolume = maxsurf
437
                    else:
438
                        cell.targetVolume = int(round( ( (
439
                        minsurf-maxsurf)/maxshared)*Sij+maxsurf ) ))
440

Modification of the LengthConstraint plugin. The influence of fibroblast-fibroblast
441
contact surface on target length for each agent is implemented according to Eq.5, in the
442
following Python code:
443

class ElongationFlexSteppable(SteppableBasePy):
444
    def __init__(self, frequency=1):
445
        SteppableBasePy.__init__(self, frequency)
446
447

    def start(self):
448
        for cell in self.cell_list:
449
            if cell.type == 1:
450
                targetLength= 160
451
                self.lengthConstraintPlugin.setLengthConstraintData(
452
                cell,16,targetLength)
453
                cell.connectivityOn = True
454
455

    def step(self, mcs):
456
        for cell in self.cell_list:
457
            if (cell.type==1):
458
                neighbor_list = self.get_cell_neighbor_data_list(cell)
459
460

                for neighbor, common_surface_area in neighbor_list:
461
                    if neighbor:
462
                        Sij=float(neighbor_list.commonSurfaceAreaWithCellTypes(
463
                        cell_type_list=[1, 2]))
464
                        if Sij>maxshared:
465
                            self.lengthConstraintPlugin.setLengthConstraintData(
466

```

```
cell,16,minlength)
elif Sij==0:
    self.lengthConstraintPlugin.setLengthConstraintData(
        cell,16,maxlength)
else:
    self.lengthConstraintPlugin.setLengthConstraintData(
        cell,16,int(round( ((minlength-maxlength)/maxshared)*
            Sij+maxlength )) )
```

Results tables of sensitivity analysis, calibration and evaluation

Table 5. Detailed results of first order wound closure sensitivity on parameters with a variation of $\pm 10\%$

Parameter	θ	Sobol index [%]
$S_{ij_{\max}}$	2.0945	15.24
λ_s	-0.475	0.78
α_c	-0.2345	0.20
β_c	-0.1495	0.08
λ_m	-0.1031	0.03
$l_{\max}$	0.03	0.003
$v_{m\,x,y}$	-2.206	16.91
$s_{\max}$	4.381	66.69
$l_{\min}$	-0.0812	0.02
$s_{\min}$	-0.021	0.002
λ_l	0.0935	0.03

Table 6. Calibration numerical results by keloid fibroblast with and without CIL hypothesis

Assays	Hypothesis	% at 20h	% at 36h
CtrlB	No CIL	37.65	41.38
	CIL	39.42	44.30
M1B	No CIL	22.30	28.20
	CIL	20.29	21.21
M2B	No CIL	45.54	51.74
	CIL	36.62	42.49

Table 7. Evaluation numerical results by keloid fibroblast with and without CIL hypothesis

Assays	Hypothesis	% at 20h	% at 36h
CtrlA	No CIL	45.96	56.72
	CIL	46.14	57.90
M1A	No CIL	21.34	26.42
	CIL	20.26	21.84
M2A	No CIL	43.46	49.86
	CIL	39.79	45.84

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